E-I balance

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Spike-Train Statistics

Before we delve deeply into Equation **??**, let's first look into the statistics of spike trains, which might help us solve this equation. As we discussed in previous lectures, such as in the integrate-and-fire model, constant external current could give rise to periodic spiking pattern with the same inter-spike interval. However, is this what we have seen in a real neuron? The answer is not. The patch-clamp recording of neuronal activity *in vivo* demonstrates that the firing pattern is highly irregular. To analyze its statistics, let's introduce a critical concept in the field of computational neuroscience, the firing rate of a neuron r. Formally, we have the following mathematical definition Given that

$$\rho(t) = \sum_k \delta(t - t_k) \tag{1}$$

The firing rate is defined as

$$r(t) = \frac{1}{\Delta t} \int_t^{t+\Delta t} d\tau \langle \rho(\tau) \rangle \tag{2}$$

where $\langle ... \rangle$ denotes an average over different trials, or even different neurons with very similar firing patterns. In practice, we may even choose a specific weighting function $\omega(\tau)$, which leads to

$$r_{approx}(t) = \int_{-\infty}^{\infty} \omega(\tau) \rho(t - \tau) d\tau$$

A useful weighting function is

$$\omega(\tau) = [\alpha^2 \tau \exp(-\alpha\tau)]_+,$$

which implies that the firing rate at time t depends only on spikes fired before t. The weight function vanishes when the argument τ is negative.

With the introduction of firing rate, we could now discuss the statistics of spike-train with some mathematical rigid. The spike-train can be viewed as a point process. In general, the probability of an event occurring at any given time could depend on the entire history of preceding events. However, if the events are themselves statistically independent, we have a Poisson process. The Poisson process provides an extremely useful approximation of irregular neuronal firing, as we will see immediately.

We denote the firing rate for a homogeneous Poisson process by $r(t) = r$, because it is independent of time. For a Poisson process, the probability of firing n spike in a time interval T is given by

$$P(n) = \frac{(rT)^n}{n!} \exp(-rT), \tag{3}$$

where rT is simply the mean number of spike in the time window T. Note that for large rT, the poisson distribution approaches a gaussian distribution. The variance of a Poisson distribution,

$$\sigma_n^2 = \langle n^2 \rangle - \langle n \rangle^2 = rT$$

Thus the variance and mean of the spike count are equal. The ratio of these two quantities, $\sigma_n^2 / \langle n \rangle$ is called the Fano factor and takes the value of 1 for homogeneous Poisson process, independent of time interval T.

Another important quantity that needs to be mentioned is the interspike interval. let's denote $\tau < t_{i+1} - t_i < \tau + \Delta\tau$, that is after the ith spike, the $i+1$ spike will occur between $t_i + \tau$ and $t_i + \tau + \Delta\tau$. The probability of having no spike in an interval τ is $\exp(-r\tau)$, and the probability to have one spike to occur between $t_i + \tau$ and $t_i + \tau + \Delta\tau$ is simply $r\Delta\tau$. Thus, the probability density to have an interspike interval τ is just given by

$$p(\tau) = r \exp(-r\tau)$$

. Moreover, We can calculate the mean interspike interval

$$\langle \tau \rangle = \int_0^\infty \tau r d\tau \exp(-r\tau) \tag{4}$$

Integrating by part, we found not surprisingly, $\langle \tau \rangle = 1/r$. The variance of the interspike interval is given by $\sigma^2(\tau) = \langle \tau^2 \rangle - \langle \tau \rangle^2$.

$$\langle \tau^2 \rangle = \int_0^\infty \tau^2 r d\tau \exp(-r\tau) = \lim_{\alpha \to 1} \frac{1}{r} \frac{d^2}{d\alpha^2} \int_0^\infty d\tau \exp(-\alpha r\tau) = \frac{2}{r^2}$$

Thus, $\sigma^2(\tau) = \frac{1}{r^2}$. As mentioned in the last lecture, we can also define the coefficient of variation of the interspike interval $\mathbf{CV} = \sigma / \langle \tau \rangle$. For poisson process, we found CV=1.

When the firing rate depends on time, we could also extend the homogeneous

Poisson process to inhomogeneous Poisson process. When n spikes occurs in an interval T with $0 < t_1 < t_2 < ... < t_n < T$, the probability density is given by

$$p[t_1, t_2, ..., t_n] = \exp\left(-\int_0^T r(t)dt\right)\prod_{i=1}^{n} r(t_i) \tag{5}$$

Balanced excitation and inhibition

Experimental data suggest that the firing pattern of a neuron *invivo* mimic the poisson process. The CV of interspike interval is close to one, and the variance of spike count in a defined time window is close to the mean. Therefore, it would be interesting to understand how does such stochasticity emerge from the interaction of thousands of neurons in the neural circuit. Let us consider a toy model, in which each neuron only has two states $\sigma_i \in [0, 1]$. At any given time, the probability that a neuron would be active is p. Each postsynaptic neuron would receive inputs from N presynaptic neurons, and the total synaptic inputs at any given time is given by

$$I_s(t) = w\sum_{i=1}^{N}\sigma_i(t) \tag{6}$$

Now let us calculate the mean and variance of the synaptic inputs. The mean is simply given by $\langle I_s(t)\rangle = wNp$. The variance

$$\begin{aligned}
\mathrm{Var}(I_s) &= \langle I_s^2\rangle - \langle I_s\rangle^2 \\
&= w^2\langle\sum_{i=1}^{N}\sigma_i^2 + \sum_{j=1}^{N}\sum_{i=1,i\neq j}^{N}\sigma_i\sigma_j\rangle - w^2\langle\sum_{i=1}^{N}\sigma_i\rangle^2 \\
&= w^2\sum_{i=1}^{N}\langle\sigma_i^2\rangle + w^2\sum_{j=1}^{N}\sum_{i=1,i\neq j}^{N}\langle\sigma_i\sigma_j\rangle - w^2\sum_{i=1}^{N}\langle\sigma_i\rangle^2 - w^2\sum_{j=1}^{N}\sum_{i=1,i\neq j}^{N}\langle\sigma_i\rangle\langle\sigma_j\rangle \\
&= w^2\sum_{i=1}^{N}\left[\langle\sigma_i^2\rangle - \langle\sigma_i\rangle^2\right] + w^2\sum_{j=1}^{N}\sum_{i=1,i\neq j}^{N}\left[\langle\sigma_i\sigma_j\rangle - \langle\sigma_i\rangle\langle\sigma_j\rangle\right] \\
&= w^2N(p-p^2) + N(N-1)\text{terms of CC}
\end{aligned} \tag{7}$$

If all inputs are independent of each other, then the cross correlation terms are zero. The variance$\mathrm{Var}(I_s) = w^2N(p-p^2)$. Now let us consider the mean and variance of the total inputs as a function of the total number of inputs N. As the network is getting larger and larger, N increases, but we do not want the activity of the neurons to explode, but to have a finite mean and variance. One way to do this is to scale the synaptic weight as $w = J/N$. In this case, $\langle I_s(t)\rangle = Jp$, and $\mathrm{Var}(I_s) = J^2(p-p^2)/N \to 0$. Basically, in a large network with independent synaptic inputs, the variance would go to zero, and all neurons should fire periodically!

How can we resolve this paradox between experimental data and the theoretical argument, which is basically the application of central limit theorem? Well, one solution is the so-called balanced excitation and inhibition. The total synaptic inputs to a neuron is the sum of excitation and inhibition. In other words,

$$\begin{aligned} \langle I_s(t) \rangle &= w_e N p_e - w_i N p_i \\ \mathrm{Var}(I_s) &= w_e^2 N(p_e - p_e^2) + w_i^2 N(p_i - p_i^2) \end{aligned} \tag{8}$$

To achieve finite variance that is independent of N, let $w_e = J_e/\sqrt{N}$ and $w_i = J_i/\sqrt{N}$, and thus $\mathrm{Var}(I_s) = J_e^2(p_e - p_e^2) + J_i^2(p_i - p_i^2)$, and $\langle I_s(t) \rangle = \sqrt{N}(J_e p_e - J_i p_i)$. Now we impose a balanced condition, $J_e = J_i p_i / p_e$, which means on average the inhibitory inputs completely cancel out the excitatory inputs.